Causality and Randomized Experiments - Group Project

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1. Introduction

As global awareness of environmental issues grows, understanding how travelers make decisions that balance cost, convenience, and sustainability has become increasingly important.

This study explores how individuals respond to information about the carbon footprint of flights, particularly focusing on the impact of how this information is presented. The central research question guiding this project is: Does the framing of carbon emissions — whether as raw numerical data or as humanized analogies — affect a traveler's willingness to pay more for lower-emission flights?

We hypothesize that humanized representations of carbon data make environmental impacts more relatable and may encourage consumers to opt for greener travel options. By analyzing survey responses from a flight booking scenario, this study aims to uncover the psychological factors that influence eco-conscious travel behaviors through the formally stated hypothesis below.

- (Null Hypothesis): There is no difference in the mean willingness to pay (WTP) for lower-emission flights between the control group and the treatment group.

$$H_0 : \mu_{\text{control}} = \mu_{\text{treatment}}$$

- (Alternative Hypothesis): There is a difference in the mean willingness to pay (WTP) for lower-emission flights between the control group and the treatment group.

$$H_a : \mu_{\text{control}} \neq \mu_{\text{treatment}}$$

1.1. Business Case

The aviation industry is a significant source of carbon dioxide (CO2) emissions, contributing approximately 2–3% of global CO2 output (Lee et al., 2021). With air travel expected to expand in the coming decades, the sector's share of global emissions is projected to rise substantially (Lee et al., 2009). Achieving global net-zero emissions and limiting climate change to 1.5°C will require considerable efforts to reduce aviation-related emissions. However, decarbonizing this sector poses substantial challenges due to its dependence on high energy-density fuels that currently lack practical alternatives (Bergero et al., 2023). In the absence of comprehensive regulatory measures, much of the responsibility for reducing emissions has fallen on voluntary initiatives by airlines and consumers (Gössling et al., 2021).

One promising approach to mitigating emissions involves influencing traveler behavior. Airlines and travel platforms have begun integrating carbon emissions data into booking systems to nudge consumers toward more sustainable options. For instance, since 2021, Google Flights has displayed carbon emissions data for individual flights, and by 2024, this information had been included in over 65 billion searches via Travalyst-affiliated platforms (Travalyst, 2024).

If presenting emissions data in an accessible format leads travelers to favor lower-emission flights, it could drive demand-side pressure on airlines to invest in cleaner technologies. Evaluating the effectiveness of these interventions hinges on understanding consumers' willingness to pay for lower-emission flights. By assessing how different framings of emissions information affect purchasing behavior, this research can inform airlines and policymakers on how to better promote sustainable air travel.

1.2. Literature Review

The existing body of research underscores the importance of message framing in shaping consumer decisions, particularly in contexts involving ethical consumption. Studies such as Smith et al. (2021) demonstrate that purely numerical data about carbon emissions often lacks emotional resonance with consumers, whereas humanized analogies—like equating emissions to the number of trees needed to absorb the CO2—can make

environmental impacts more tangible. Johnson and Lee (2020) similarly found that framing environmental data in relatable ways increases the likelihood of consumers making sustainable choices.

While there is a growing literature on framing effects in areas such as energy use and food consumption, fewer studies have examined how these dynamics influence travel decisions. Existing research on willingness to pay for lower-emission flights (e.g., Baumeister et al., 2022; Carroll et al., 2022; Sanguinetti & Amenta, 2022; Song et al., 2024; Crosby, Thompson & Best, 2024) often employs discrete choice experiments but faces notable limitations. Many studies use non-representative samples and do not differentiate between leisure and business travelers, even though these groups may have different priorities and constraints. Furthermore, some experimental designs fail to accurately replicate real-world booking environments, raising concerns about hypothetical bias and external validity.

This study seeks to address these gaps by using a more realistic choice experiment modeled after popular booking interfaces, such as Google Flights, and by separately analyzing the responses. This approach allows for a more nuanced understanding of how different groups perceive emissions data and how framing affects their willingness to pay for greener flights. Ultimately, this research contributes to the broader discourse on sustainable consumption and behavioral economics, offering practical insights for airlines and policymakers aiming to reduce the aviation sector's carbon footprint.

2. Experimental Design

To investigate whether the framing of carbon emissions affects travelers' willingness to pay (WTP) for lower-emission flights, we implemented a completely randomized experimental design. In this approach, participants were divided into treated and control groups in the form of a survey in an A/B setting created on the platform Qualtrics. In this sense, participants were randomly assigned to one of two conditions: one group, (control), received carbon emissions information in raw numerical terms (e.g., kilograms of CO2), while the other group, (treatment), was presented with "humanized" analogies (e.g., the equivalent number of football fields).

This approach ensures that a fixed number of subjects receive the treatment, with assignments made randomly to balance the groups as much as possible. While the number of treated and control units does not have to be equal, it is predetermined to ensure sufficient representation in both groups. Complete randomization does not stratify participants based on predefined variables; this method simplifies the experimental design and allows for broader generalizability of results.

While a completely randomized design has its strengths, it is not without limitations. The random assignment of participants to different conditions may still introduce variability due to individual differences that are not controlled for. Additionally, common survey-related challenges such as response bias could affect the validity of our findings. Nonetheless, this design should allow us to make meaningful conclusions about how the framing of carbon emissions influences travelers' decisions.

Despite this limitation, complete randomization remains a widely used approach for estimating causal effects, ensuring that differences in responses are attributable to the treatment rather than systematic pre-existing differences between participants.

When designing the survey, we considered several potential challenges to ensure the reliability and generalizability of our results. One of the primary concerns was the risk of a homogenous respondent pool in terms of demographics such as age, nationality, or occupation, which could limit the generalizability of our findings. To address this, we made efforts to recruit participants from diverse geographical locations and encouraged participants, especially family and friends, to share the survey within their networks to reach varied age groups and backgrounds.

We also acknowledged the limited attention span of participants and aimed to minimize survey fatigue. To mitigate this, we designed the survey to be as concise as possible, using clear and straightforward language. The questions were structured to be easy to understand, and the answer options were kept simple to maximize response rates, ensure comparability, and facilitate subsequent data analysis.

Another consideration was the framing of carbon emission information in the survey. We aimed to avoid introducing bias that could stem from participants' prior associations or experiences. For instance, when presenting humanized framing examples, such as comparing carbon emissions to the number of football fields, we ensured that these comparisons were neutral and universally understandable, avoiding culturally specific references that might influence participants' responses.

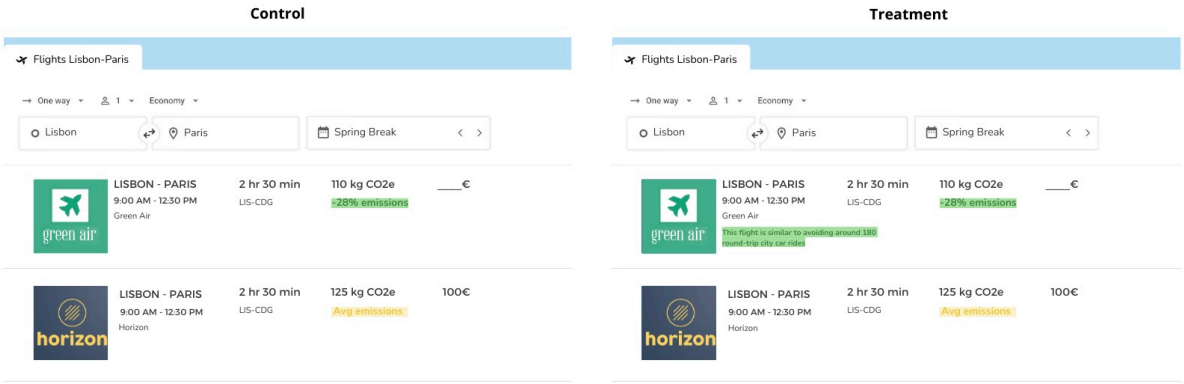


Figure 1: Images displayed for Control and Treatment Groups in Survey

From an ethical standpoint, we prioritized participant confidentiality and voluntary participation. A confidentiality statement was included at the beginning of the survey, informing participants that their responses would remain anonymous, data would be kept confidential, and participation was entirely voluntary.

The complete survey design and materials can be found in the appendix.

To rigorously evaluate the causal effect of framing on WTP, we specify the following baseline regression model:

$$WTP_i = \beta_0 + \beta_1 \text{Humanized}_i + \epsilon_i$$

Where WTP_i represents the willingness to pay for individual i and Humanized_i is a binary variable indicating whether the participant received the humanized framing. The coefficient β_1 captures the direct impact of framing on WTP.

Since treatment effects may vary across individuals, we extend the model to account for heterogeneous treatment effects by incorporating key demographic covariates :

$$WTP_i = \beta_0 + \beta_1 \text{Humanized}_i + \beta_2 \text{Gender}_i + \beta_3 \text{Age}_i + \beta_4 \text{Income}_i + \beta_4 \text{Nationality}_i + \epsilon_i$$

By including these additional covariates, we control for potential confounding factors and explore whether certain subgroups (e.g., frequent travelers) exhibit stronger responses to the treatment. This approach allows us to better understand the mechanisms through which framing influences WTP.

To determine the appropriate sample size for detecting a meaningful effect, we conducted a power analysis for a one-sided two-sample t-test. Given a small effect size (d = 0.2), a significance level of $\alpha = 0.05$, and a power of 0.8, the required sample size was calculated using the `power.t.test` function in R. The analysis indicated that the sample should contain approximately 256 participants to ensure the desired statistical power to detect the expected effect.

Table 1: Sample Size

Group	Results
Estimated n	256
Delta	0.220
Significance Level	0.050
Power	0.800
True n	258

3. Analysis and Results

To better understand the sample data, the following section provides a comprehensive analysis of the characteristics and distributions of the collected variables.

3.1. Data Description

The survey, conducted between March 16 and March 22, 2025, yielded 208 usable responses. We also considered 50 more entries from Varteo's synthetic survey, reaching the final size of 258 responses; while 258 responses provide a reasonable basis for statistical analysis, the sample may still be considered moderate rather than large, which could limit the robustness of subgroup analyses. A larger sample would increase statistical power and allow for more precise estimation of treatment effects.

To ensure a clean and structured dataset for analysis, we first loaded the survey data and removed unnecessary rows and columns, including automatically generated metadata from Qualtrics. We then filtered out incomplete responses and those from preview distributions to maintain data integrity. Next, we renamed variables for consistency and ease of use in R. A treatment group variable was created based on the 'ImageShown' column, classifying participants accordingly. We proceeded by converting categorical variables related to environmental and ethical attitudes into numerical scales, facilitating quantitative analysis. To enhance interpretability, we computed mean scores for environmental and ethical perceptions, aggregating responses across multiple related survey items. These preprocessing steps ensured that our dataset was well-structured, reliable, and ready for statistical analysis.

Table 2: Overview and Description of Variables

Variable	Description	Format	Collection
flight_frequency	Flight frequency of the participant	Factor	Drop-down menu
nationality	Nationality of the participant	Factor	Drop-down menu
gender	Gender of the participant	Factor	Drop-down menu
age	Binned age of the participant	Factor	Drop-down menu
work_status	Profession of the participant	Factor	Drop-down menu
education_level	Education level of the participant	Factor	Drop-down menu
household_income	Household income level of the participant	Factor	Drop-down menu
group	Indicating assigned treatment or control group	Factor	Automatically
env_mean	Environmental mean score of the participant	Numeric	5-point scale
eth_mean	Ethical mean score of the participant	Numeric	5-point scale
wtp	Willingness to pay	Numeric	Slider

Our primary outcome variable was WTP, which measured participants' willingness to pay for a lower-emission flight under different treatment or control conditions. Participants provided their responses via a slider input, allowing for precise and unrestricted values. Given the potential influence of environmental

and ethical attitudes on willingness to pay, we computed mean scores for both constructs using a 5-point Likert scale. The flight frequency of each participant was recorded to assess their typical travel behavior. Demographic variables such as nationality, gender, age, work status, education level, and household income were collected using drop-down menus to ensure consistency in responses. Lastly, participants were randomly assigned to either the control or treatment group, which was recorded in the group variable for experimental analysis.

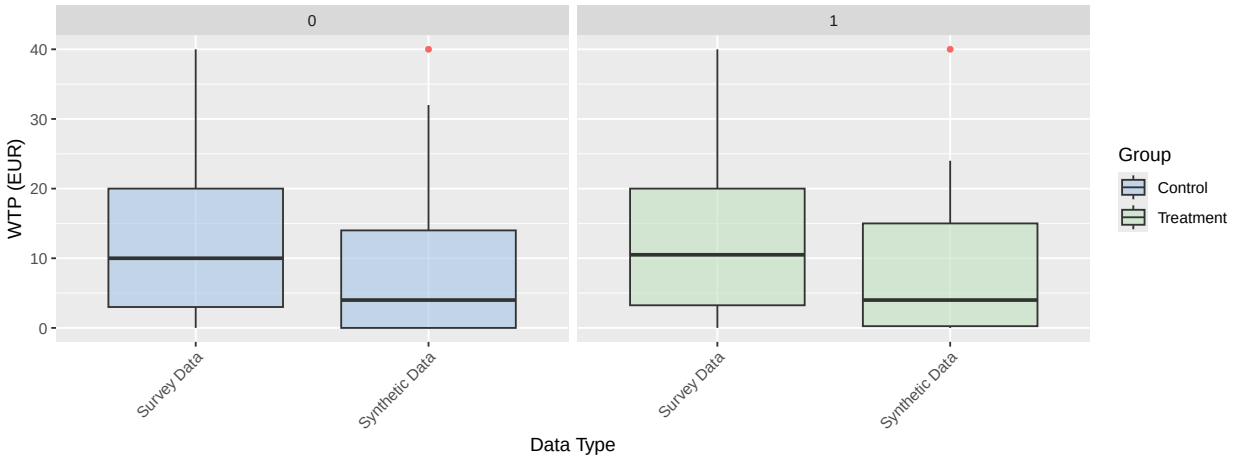
The boxplot in Graph 1 below compares the Willingness to Pay (WTP) in EUR between data collected through traditional Survey Data and Synthetic Data generated via Varteo, across two groups: Control and Treatment. Varteo employs a sophisticated algorithm that ensures synthetic personas make decisions consistent with real individuals by continuously monitoring their decision-making processes against real-world data. If a synthetic persona deviates significantly from human behavior, it is flagged for review to maintain the model's accuracy.

From the graph, we observe that the Survey Data and Synthetic Data exhibit similar median WTP values across both Control and Treatment groups. However, the Synthetic Data shows a slightly more compressed distribution, suggesting that Varteo's modeling process reduces extreme variations compared to Survey Data. The Survey Data displays a wider interquartile range (IQR) and more pronounced outliers, particularly in the higher WTP values. This indicates that responses from real individuals may be more variable and prone to extreme valuations, whereas Varteo's personas produce more stabilized predictions.

The comparison between synthetic and survey data in Appendix A reveals some differences across key covariates. The synthetic data has a lower proportion of respondents aged 18-24 (12% vs. 39%) and fewer females (48% vs. 60%) compared to the survey data. Additionally, the proportion of individuals with a bachelor's degree is higher in the synthetic data (76% vs. 43%), as is the proportion of employed respondents (82% vs. 56%) and Portuguese nationals (98% vs. 82%). Importantly, Willingness to Pay (WTP) is lower in the synthetic data (9.56 EUR vs. 12.87 EUR).

Considering these results, the following analysis will integrate Varteo's data as a complementary source for real survey data, ensuring robustness while maintaining real-world variability.

Graph 1: WTP Boxplot by Group and Data Type



It is important to mention that concerns related to non-compliance, attrition, and interference were minimal. Non-compliance was not an issue since treatment assignment closely aligned with treatment receipt, ensuring that all participants followed their assigned conditions. Attrition was also not a major concern, as outcome data was collected for nearly all respondents, minimizing the risk of bias due to missing observations. Finally, interference was unlikely because participants completed the survey independently, without exposure to or influence from others' treatment assignments. These factors help ensure that the estimated effects accurately reflect the intended causal relationships.

3.2. Exploratory Data Analysis

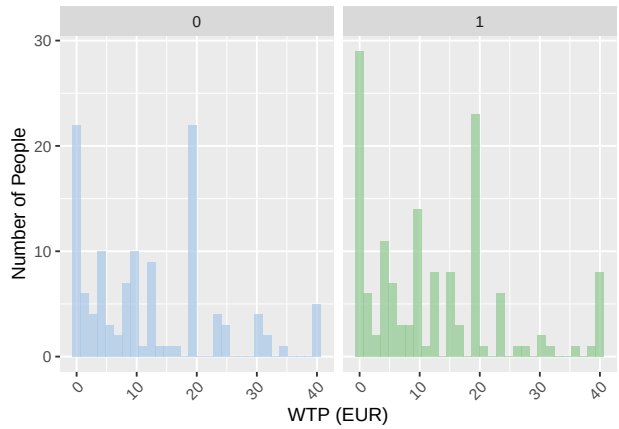
Outcome Variable - Willingness to Pay

To begin our Exploratory Data Analysis, we first examine how our outcome variable is distributed across groups.

The histogram below shows the distribution of Willingness to Pay (in €). In both groups, the most common values chosen were 0 and 20€. While the majority of participants selected price points between 0 and 20€, the treatment group showed a slight preference for price points between 20 and 40€. Notably, the distribution is not bell-shaped, indicating that a non-parametric test may be more appropriate for further analysis.

WTP Distribution and Summary Statistics

Graph 2: WTP Histogram by Group

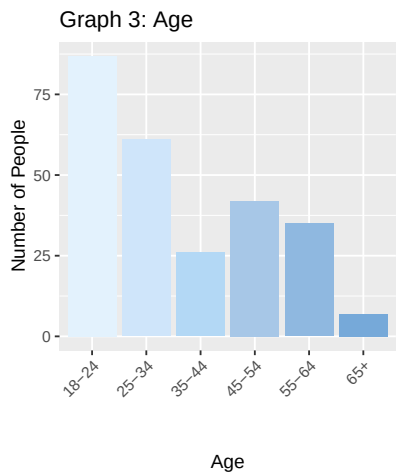


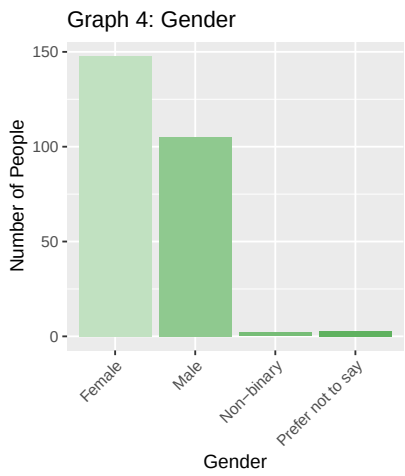
Statistic	Group	Value
Min	Control	0.00
	Treatment	0.00
Median	Control	10.00
	Treatment	10.00
Mean	Control	12.16
	Treatment	12.29
Max	Control	40.00
	Treatment	40.00

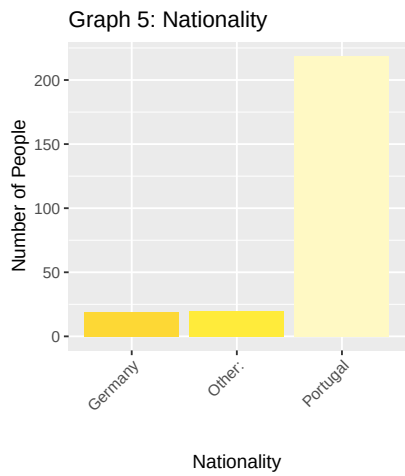
Main Covariates

While examining the distribution of our outcome variable (WTP) provides insights into its overall behavior, it is equally important to explore the distribution of key covariates. These variables may influence WTP and help us understand whether differences across groups arise from the treatment itself or pre-existing characteristics.

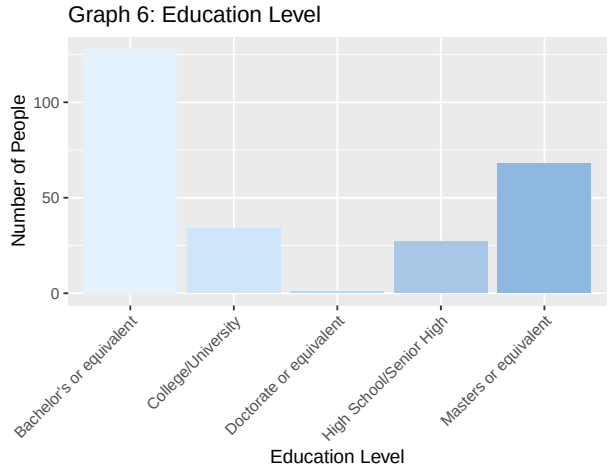
As can be seen from the far left graph below, approximately one third of the participants are aged between 18-24 (n = 87). There are significantly more female participants than male (n = 148 vs n = 105). Most of the participants were of Portuguese nationality (n = 219), followed by German (n = 19). Other different nationalities were found to be in the minority and were grouped together for visualization purposes (n = 20).

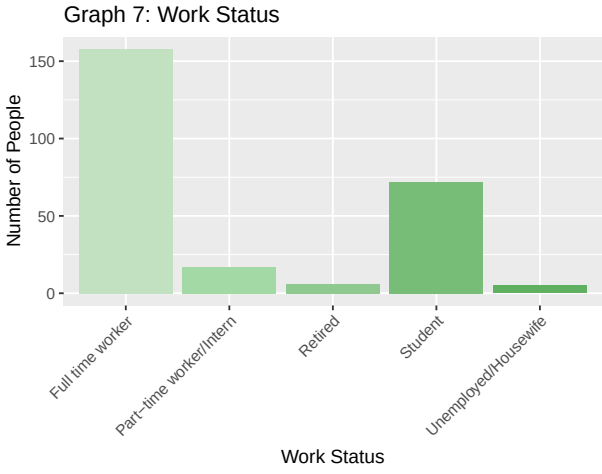






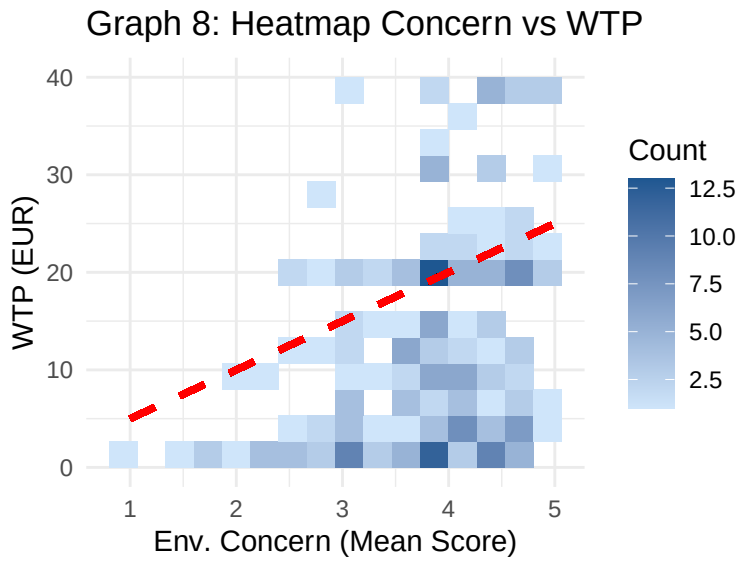
The survey also gathered data on participants' education levels, revealing that the majority held a Bachelor's degree (n = 128), followed by those with a Master's degree (n = 68). A smaller portion of respondents had a Doctor's degree (n = 1). Regarding employment status, the sample was largely composed of full-time workers (n = 158) and students (n = 72).





Graph 9 reveals an interesting trend: individuals who exhibit higher levels of environmental concern, as measured through a series of survey questions assessing their attitudes toward environmental issues, tend to have a greater willingness to pay (WTP). This suggests a positive relationship between environmental awareness and financial commitment to environmentally friendly initiatives.

While this trend aligns with expectations, it is important to assess whether this relationship holds when controlling for other factors. In the next section, we further explore the role of covariates and treatment effects through regression analysis to determine whether environmental concern remains a significant predictor of WTP.



- **Balance Check**

The likelihood ratio test below compares two models to assess whether adding predictors improves model fit. Model 1 includes only an intercept, while Model 2 includes age, nationality, education level, gender, work status, and household income as predictors. The test yields a chi-square statistic of 38.671 with 22 degrees of freedom and a p-value of 0.01539, indicating a significant difference between the models. In the context of a balance check, this suggests that these covariates are not evenly distributed across groups — if they were, adding them would not significantly improve the model. The significant result implies potential imbalance, meaning these factors might systematically differ between groups.

The logistic regression analysis suggests that household income significantly predicts treatment assignment, with higher-income participants being more likely to be in the treatment group. Additionally, the covariate balance table, (Appendix A), highlights significant differences in nationality ($p = 0.038$) and work status ($p = 0.021$), with Portuguese respondents and non-full-time workers being overrepresented in the treatment group. Although other variables, such as education level and gender, appear balanced, the observed imbalances in nationality, work status, and income could introduce bias. To address this, adjustments using regression models will be considered in subsequent analyses to ensure robust causal inference.

Table 3: Likelihood Ratio Test Results

	#Df	LogLik	Df	Chisq	Pr(>Chisq)
1	1	-177.893			
2	23	-158.557	22	38.671	0.015

- **Correlation Matrix**

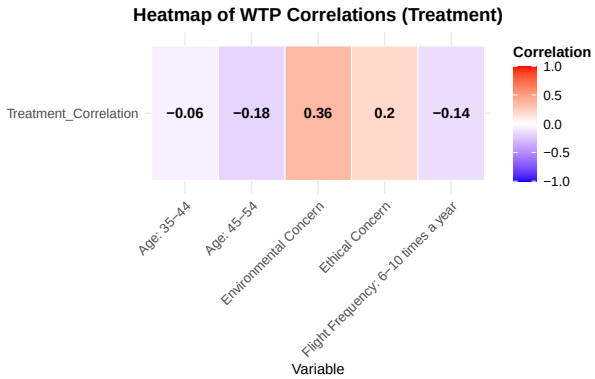
Before proceeding with our regression models and subsequent tests, we first decided we should examine the correlation between our variables and willingness to pay (WTP).

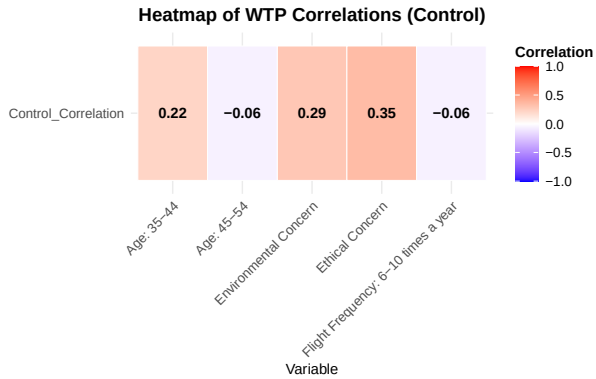
The correlations are generally low, with all absolute values below 0.5. This suggests that no single factor strongly drives WTP on its own. However, the sign and significance of these correlations can still provide meaningful insights.

Within the treatment group, there is a positive correlation between environmental concern (env_mean) and WTP, meaning individuals who care more about environmental issues are willing to pay more. Conversely, ethical concern (eth_mean) shows no significant correlation with WTP. This suggests that environmental concerns might have a stronger influence on WTP than ethical values in the treatment group.

Age shows a negative correlation with WTP, especially in the 35-44 and 45-54 age groups; income does not appear to significantly affect WTP, as the correlation is minimal across different income categories. Similarly, flight frequency does not have a strong impact on WTP, with only the 6-10 times a year category showing a slight negative relationship, though not statistically significant.

Gender and other demographic factors were also analyzed. The correlations between these factors and WTP are low, suggesting that there are no clear patterns based solely on these characteristics.





ATE and Mean Comparison

While the correlation matrix provides insights into how different variables relate to WTP, it does not establish causality. To further investigate the effect of the treatment on WTP, we calculate the Average Treatment Effect (ATE) and compare the mean WTP between treatment and control groups. This allows us to assess whether any observed differences in WTP can be attributed to the intervention rather than underlying correlations.

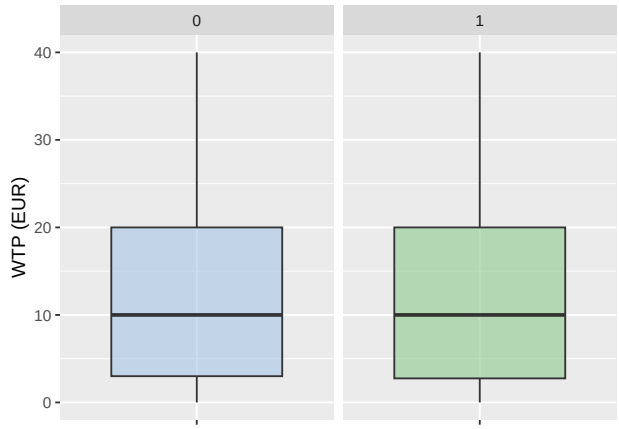
Graph 8 below compares the Willingness to Pay (WTP) between participants in the Control and Treatment groups. The distributions are fairly similar, with both groups having a median around 10€. Additionally, the interquartile range (IQR) for both groups spans approximately 5€ to 20€, indicating comparable variability in responses.

While there does not appear to be a notable difference in WTP between groups, suggesting that the treatment effect may be small, we will conduct formal statistical tests to determine whether this difference is statistically significant.

To quantify the impact of framing on participants' willingness to pay, we estimated the true Average Treatment Effect (ATE) across all participants. The results, summarized in the table below, show the mean WTP for each group:

Means of WTP by group and ATE

Graph 9: WTP Boxplot by Group



Group	Mean_WTP
Control (0)	12.16
Treatment (1)	12.29
ATE	0.13

The ATE of 0.130 suggests that the treatment increased the average WTP compared to the control group. This indicates that framing had a positive impact, though the magnitude is relatively small. To determine whether this difference is statistically significant or if it could have arisen due to random chance, we will conduct formal statistical tests in the following section.

3.3. Findings

Starting off our regression analysis, we feel it is important to first assess whether there are baseline differences between the treatment and control groups. Conducting a preliminary test, such as a Welch Two-Sample t-test, helps determine if the groups are already significantly different in terms of their willingness to pay (WTP) before accounting for additional covariates. If a significant difference were found, it would suggest that factors other than the treatment itself might be influencing WTP, highlighting the need for careful adjustment in the regression model. Conversely, if no significant difference is found, it provides reassurance that any potential effects observed in the regression analysis are more likely to be attributable to the included explanatory variables rather than pre-existing imbalances.

The table below we have the results of the Welch Two-Sample t-test, which indicate that there is no significant difference in the mean willingness to pay (WTP) between the control group (M = 12.16) and the treatment group (M = 12.29), t(250.77) = -0.09, p = 0.9287. The p-value is much higher than the conventional significance threshold of 0.05, suggesting that any observed difference is likely due to random variation rather than a true effect.

The 95% confidence interval (-3.26, 3.02) includes zero, further supporting the conclusion that the two groups do not significantly differ in their WTP for lower-emission flights. Therefore, we fail to reject the null hypothesis, indicating that the treatment did not have a measurable impact on WTP.

Table 4: Welch Two-Sample t-test Results

	Statistic	Value
1	t-value	-0.090
2	df	250.775
3	p-value	0.929
4	95% CI Lower	-2.867
5	95% CI Upper	2.618
6	Mean (Control)	12.161
7	Mean (Treatment)	12.286

After confirming that there was no significant difference in WTP between the control and treatment groups with the Welch t-test, we wanted to ensure that this test was indeed appropriate. Although the t-test is generally preferred because it compares means rather than distributions (as the Mann-Whitney U test does), it relies on the assumption of normally distributed data. Since we believe our sample size is statistically large enough for the Central Limit Theorem (CLT) to hold, this assumption may not be critical. However, to avoid potential biases and strengthen our conclusions, we decided to formally test for normality.

The results of the Shapiro-Wilk test indicate that the assumption of normality is violated for both the control ($W = 0.897$, $p < 0.001$) and treatment ($W = 0.888$, $p < 0.001$) groups. Since the p-values are significantly below 0.05, we reject the null hypothesis of normality, suggesting that the distribution of WTP (Willingness to Pay) is not normally distributed in either group.

Table 5: Shapiro-Wilk Test Results

	Group	W	p_value
1	Control	0.897	1.67190e-07
2	Treatment	0.888	7.14342e-09

Given this result, we could still justify using the t-test under the assumption that our sample size is sufficiently large for the CLT to apply. However, to ensure robustness, we performed an additional non-parametric test (Mann-Whitney U test), which does not assume normality. The resulting p-value (0.9785) is considerably high (greater than 0.05), indicating that we fail to reject the null hypothesis. This suggests strong evidence that the distribution of WTP is statistically similar between the groups, implying that the treatment had no significant effect on participants' willingness to pay.

Since both tests led to the same conclusion—that the treatment had no effect on WTP—we can confidently assert that our findings are not driven by assumptions about normality and that the treatment did not have a measurable impact on WTP.

Table 6: Mann-Whitney U Test Results

Statistic	P_Value
$8,276.5000$	0.9785

Although our previous tests indicated that the treatment likely had no effect on WTP, we decided to further explore this relationship by controlling for additional covariates. By doing so, we aimed to determine whether any underlying factors influenced WTP and to confirm that the treatment remained statistically insignificant even when accounting for these variables. To achieve this, we implemented a baseline model (including only treatment) and an expanded model incorporating covariates and interaction effects.

We conducted an Ordinary Least Squares (OLS) regression analysis to assess the relationship between the treatment and other independent variables with WTP. The regression model helps identify the impact of each factor while accounting for the influence of others. We considered several different models with varying sets of covariates.

The baseline model serves as the baseline, including only the treatment variable (group). The coefficient for group ($\beta = 0.125$) is small and statistically insignificant, indicating that the treatment had no observable impact on WTP. The R^2 value is nearly zero, meaning that the model explains virtually none of the variation in WTP. This suggests that simply receiving the treatment is not enough to predict changes in WTP.

The extended model further expands the analysis by adding nationality and interaction effects. A notable new finding is that Portuguese participants exhibit significantly lower WTP ($\beta = -8.970$, $p < 0.05$), suggesting that nationality plays a crucial role in willingness to pay.

The R^2 for the extended model improves to 0.239, meaning that 23.9% of the variance in WTP is explained by the included covariates. However, the treatment variable (group) remains statistically insignificant, reinforcing the conclusion that the intervention did not meaningfully influence participants' WTP.

Models 1 and 4 are displayed in the table below, while the other evaluated models can be found in Appendix A.

Table 7: Regression Results for WTP Models

	Dependent variable:	
	WTP	
	Baseline Model	Extended Model
	(1)	(2)
group	0.125 (1.398)	0.953 (1.437)
flight_frequency6-10 times a year		-4.789^{*} (2.902)
flight_frequencyMore than 10 times a year		−0.015 (3.670)
flight_frequencyNever		3.197 (2.670)
flight_frequencyOnce a year or less		1.475 (1.664)
household_income1500 euros to 4999 euros per month		−0.038 (6.123)
household_income5000 euros to 7999 euros per month		−2.137 (6.274)
household_income8000 to 11999 euros per month		−2.706 (6.982)
household_incomeLess than 1500 euros per month		−3.898 (6.354)
env_mean		3.724^{***} (1.326)
eth_mean		2.786 (2.038)
age25-34		−3.433 (2.412)
age35-44		−0.888 (3.345)
age45-54		-7.431^{***} (2.653)
age55-64		−1.726 (3.221)
age65+		−5.336 (8.433)
genderMale		1.213 (1.435)
genderNon-binary		2.472 (28.968)
genderPrefer not to say		0.053 (5.145)
nationalityOther:		−5.756 (4.059)
nationalityPortugal		-8.970^{**} (3.655)
work_statusPart-time worker/Intern		1.183 (2.961)
work_statusRetired		6.261 (10.688)
work_statusStudent		−1.529 (2.109)
work_statusUnemployed/Housewife		−2.546 (8.851)
Constant	12.161^{***} (1.017)	−3.101 (8.274)
Observations	258	255
R^2	0.00003	0.239
Adjusted R^2	−0.004	0.156
F Statistic	0.008 (df = 1; 256)	2.883^{***} (df = 25; 229)
Note:		*p<0.1; **p<0.05; ***p<0.01

4. Discussion

4.1. Final Considerations

Internally, the study benefits from a completely randomized experimental design, which enhances causal inference by reducing systematic differences between the treatment and control groups. The inclusion of synthetic data from Varteo also contributes to the robustness of the findings by offering a supplementary dataset that closely mirrors real-world decision-making patterns.

Externally, while the sample includes participants from diverse geographical backgrounds, it is still skewed toward Portuguese respondents, potentially limiting the generalizability of the findings. Additionally, the survey setting differs from real-world flight booking scenarios, where consumers face additional constraints, such as time pressure and brand preferences.

The results show that the humanized framing of carbon emissions did not significantly increase willingness to pay (WTP) for lower-emission flights. While this does not negate the psychological impact of framing, it suggests that WTP might be driven more by personal values and economic considerations than by information presentation alone.

Another important factor at play was that the assumption of normality was violated in both treatment and control groups, as indicated by the Shapiro-Wilk test. This necessitated the use of non-parametric tests, which may have reduced the power of the statistical analysis.

Some additional limitations of this study should be acknowledged. First, the sample is not fully representative of the global traveling population, as Portuguese participants constitute a large majority. This demographic imbalance may introduce cultural biases that affect the generalizability of the findings, particularly regarding attitudes toward sustainability and willingness to pay (WTP) for lower-emission flights.

Second, the reliance on self-reported data presents inherent challenges. Respondents may overestimate or underestimate their WTP due to the absence of real financial stakes in the decision-making process. Hypothetical bias remains a concern, as individuals might express a willingness to pay more for greener options in a survey setting but behave differently in real-world purchase scenarios.

Finally, the experimental treatment was administered in a one-time survey, whereas real consumer behavior is often shaped by repeated exposure to sustainability messaging over time. In practice, travelers may develop stronger or weaker preferences for lower-emission flights through cumulative interactions with environmental campaigns, airline policies, or social influences.

4.2. Summary and Recommendations

The results indicate that the humanized framing of carbon emissions did not produce a statistically significant increase in willingness to pay (WTP) for lower-emission flights. The mean WTP difference between the treatment and control groups was negligible, and the regression results confirm the absence of a significant effect ($p > 0.05$). This suggests that, while framing may influence consumer perceptions, it does not necessarily translate into higher financial commitments toward sustainable travel.

A statistically significant relationship was observed between environmental concern scores and WTP, independent of the framing condition. Individuals with higher self-reported environmental concern exhibited a greater willingness to pay for greener flights, reinforcing the idea that pre-existing environmental values play a stronger role in decision-making than subtle variations in information presentation.

Frequent flyers (6–10 trips per year) demonstrated a lower WTP for sustainable flight options compared to infrequent travelers. Regression analysis confirms a statistically significant negative effect. This may be attributed to cost sensitivity or established habitual behaviors, which make price a dominant factor in their decision-making.

While broad demographic variables such as income and education did not exhibit statistically significant direct effects on WTP, Portuguese participants showed a notably lower WTP relative to other nationalities. This finding highlights potential cultural influences that shape attitudes toward sustainable travel and suggests that country-specific factors, such as environmental policies and economic conditions, may mediate consumer behavior.

To better understand how framing affects behavior, future studies should test these effects in actual booking environments where real financial commitments are made. Examining how repeated exposure to sustainability messaging affects WTP over time could also provide more insights into the long-term impact of framing.

Exploring different types of humanized framing could be a potential source of new findings, such as social norm messaging (e.g., "80% of travelers choose greener flights") or financial incentives tied to emissions reduction. Given the observed differences in WTP based on nationality, further studies should investigate how cultural norms and economic conditions influence sustainable travel decisions.

Overall, while this study does not find strong evidence that humanized framing alone significantly increases WTP, it provides valuable insights into the factors that drive sustainable travel decisions. These findings can inform more effective strategies for promoting eco-friendly travel choices in the aviation industry.

5. AI Disclaimer

AI was used for the creation of this report in a limited scope. The main use case of ChatGPT was the debugging of code. Furthermore, minor paragraphs were rephrased.

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Appendix A

Table 8: Comparison of Covariates Between Synthetic and Survey Data

Covariate	Survey Data	Synthetic Data
DistributionChannel	Survey Data	Synthetic Data
Age_18_24	0.39	0.12
Gender_Female	0.60	0.48
Educ_Bachelor	0.43	0.76
Work_Employed	0.56	0.82
Nat_Portugal	0.82	0.98
WTP	12.87	9.56

Table 9: Proportion of Key Covariates by Group in Synthetic and Survey Data

Covariate	Control	Treatment
group	0	1
Age_18_24_Survey Data	0.270	0.490
Age_18_24_Synthetic Data	0.170	0.080
Gender_Female_Survey Data	0.630	0.570
Gender_Female_Synthetic Data	0.460	0.500
Educ_Bachelor_Survey Data	0.450	0.420
Educ_Bachelor_Synthetic Data	0.750	0.770
Work_Employed_Survey Data	0.680	0.460
Work_Employed_Synthetic Data	0.790	0.850
Nat_Portugal_Survey Data	0.740	0.880
Nat_Portugal_Synthetic Data	0.960	1
WTP_Survey Data	12.670	13.040
WTP_Synthetic Data	10.170	9

Table 10: Covariates table: Balance check

	Characteristic	Category	Control	Treatment	p-value
1		age			NA
2		18-24	29 (33%)	58 (67%)	NA
3		25-34	27 (44%)	34 (56%)	NA
4		35-44	18 (69%)	8 (31%)	NA
5		45-54	19 (45%)	23 (55%)	NA
6		55-64	22 (63%)	13 (37%)	NA
7		65+	3 (43%)	4 (57%)	NA
8	nationality	nationality			0.0383
9		Germany	13 (68%)	6 (32%)	NA
10		Other:	12 (60%)	8 (40%)	NA
11		Portugal	93 (42%)	126 (58%)	NA
12	education_level	education_level			0.3488
13		Bachelor's or equivalent	60 (47%)	68 (53%)	NA
14		College/University	16 (47%)	18 (53%)	NA
15		Doctorate or equivalent	1 (100%)	0 (0%)	NA
16		High School/Senior High	8 (30%)	19 (70%)	NA
17		Masters or equivalent	33 (49%)	35 (51%)	NA
18	gender	gender			0.6056
19		Female	70 (47%)	78 (53%)	NA
20		Male	46 (44%)	59 (56%)	NA
21		Non-binary	0 (0%)	2 (100%)	NA
22		Prefer not to say	2 (67%)	1 (33%)	NA
23	work_status	work_status			0.0214
24		Full time worker	83 (53%)	75 (47%)	NA
25		Part-time worker/Intern	9 (53%)	8 (47%)	NA
26		Retired	1 (17%)	5 (83%)	NA
27		Student	24 (33%)	48 (67%)	NA
28		Unemployed/Housewife	1 (20%)	4 (80%)	NA
29	household_income	household_income			0.329
30		12000+ euros per month	6 (86%)	1 (14%)	NA
31		1500 euros to 4999 euros per month	65 (45%)	81 (55%)	NA
32		5000 euros to 7999 euros per month	19 (46%)	22 (54%)	NA
33		8000 to 11999 euros per month	7 (41%)	10 (59%)	NA
34		Less than 1500 euros per month	21 (45%)	26 (55%)	NA

Table 11: Other Regression Results for WTP Models

	Dependent variable:	
	Willingness to Pay (WTP)	
	Model 2	Model 3
	(1)	(2)
group	0.548	0.505
	(1.372)	(1.410)
flight_frequency6-10 times a year	−4.626*	−4.938*
	(2.520)	(2.615)
flight_frequencyMore than 10 times a year	−0.956	0.920
	(3.495)	(3.345)
flight_frequencyNever	2.201	2.597
	(2.408)	(2.516)
flight_frequencyOnce a year or less	0.924	1.187
	(1.591)	(1.604)
household_income1500 euros to 4999 euros per month	−2.245	−0.539
	(4.405)	(4.316)
household_income5000 euros to 7999 euros per month	−3.699	−2.102
	(4.516)	(4.503)
household_income8000 to 11999 euros per month	−5.058	−3.468
	(4.969)	(5.141)
household_incomeLess than 1500 euros per month	−4.989	−3.809
	(4.542)	(4.438)
env_mean	4.262***	4.314***
	(1.232)	(1.249)
eth_mean	1.490	2.114
	(1.717)	(1.825)
age25-34		−0.610
		(1.772)
age35-44		0.339
		(2.999)
age45-54		−6.478***
		(1.909)
age55-64		0.710
		(2.710)
age65+		1.869
		(4.706)
genderMale		1.410
		(1.363)
genderNon-binary		2.761
		(7.809)

Appendix B

English

Intro

Welcome!

Thank you for taking the time to participate in this survey. We're interested in learning more about how people make travel decisions, their views on ethical consumption, and environmental attitudes. We'll start with a few questions about your flying habits.

Later, you'll be asked to consider a flight booking scenario before finishing with some brief demographic questions. This should take about 5 minutes, and there are no right or wrong answers—just your honest opinions.

Your responses will remain completely anonymous, and all data collected will be kept strictly confidential and used solely for research purposes. Participation in this survey is entirely voluntary, and you may choose to stop at any time without any consequences. By proceeding, you acknowledge that you have read and understood this statement and agree to participate voluntarily.

Let's get started!

Flying

How often do you fly per year?

- Never
- Once a year or less
- 2-5 times a year
- 6-10 times a year
- More than 10 times a year

On average, how long is the duration of your typical flight?

- Less than 1 hour

- 1-2 hours
- 2-4 hours
- 4-6 hours
- More than 6 hours

Please rank the following factors in order of importance when choosing a flight, with 1 being the most important and 5 being the least important.

Lower price

Direct route

Airline

More sustainable practices

Free in-flight meals

Rank the following perks based on how willing you are to pay extra for them, with 1 being the perk you are most willing to pay for and higher numbers being less important.

Extra baggage

Priority boarding

More spacious seating

Faster security screening

Sustainability practices

Ethical & Environmental

Please indicate how much you agree or disagree with each statement.

	Strongly Agree	Agree	Neither Agree nor Disagree	Diasagree	Strongly Disagree
I consider ethical and environmental factors when making everyday purchasing decisions.	○	○	○	○	○
I make an effort to reduce waste, such as limiting single-use plastics or minimizing unnecessary travel.	○	○	○	○	○
I am willing to pay more for services that are environmentally friendly.	○	○	○	○	○
I avoid companies that are known for environmentally harmful practices.	○	○	○	○	○
I believe businesses should take more responsibility for reducing their environmental impact.	○	○	○	○	○

Please indicate how much you agree or disagree with each statement.

	Strongly Agree	Agree	Neither Agree nor Disagree	Disagree	Strongly Disagree
I make an effort to support brands that align with my ethical values.	○	○	○	○	○
I consider ethical factors when deciding which businesses to support.	○	○	○	○	○
I am willing to change my purchasing habits to support ethical or sustainable businesses.	○	○	○	○	○
I believe that ethical consumer choices can make a real impact on social and environmental issues.	○	○	○	○	○

	Strongly Agree	Agree	Neither Agree nor Disagree	Disagree	Strongly Disagree
I agree with behavioral nudges (such as labeling or incentives) to encourage more ethical and sustainable choices.	○	○	○	○	○

Flight Context

Flight Booking Scenario:

Spring break is coming up, and you're ready for a well-deserved getaway. After considering several destinations, you decide on **Paris**—a city you've always enjoyed visiting. With your travel dates in mind, you open your laptop and start searching for flights.

As you search, you come across a variety of options—different airlines, flight durations, layovers, and prices. Some tickets seem like a great deal but come with long stopovers, while others offer more comfort at a higher cost. With so many factors to consider, you start weighing what matters most to you when booking a flight.

After searching for some time, **you narrow it down to two flights that fit your needs well**.

Flight

If the Horizon flight serves as a baseline, how much more, if anything, would you be willing to pay for a flight with lower emissions?

	0	4	8	12	16	20	24	28	32	36	40
I would pay...											☐

Demographic

Which country are you from

- ○ Germany
- ○ Portugal
- ○ ☐ Other:

What is your gender?

- Male
- Female
- Non-binary
- Prefer not to say

What is your age?

- 18-24
- 25-34
- 35-44
- 45-54
- 55-64
- 65+

What is your current education level?

- High School/Senior High
- College/University
- Bachelor's or equivalent
- Master's or equivalent
- Doctorate or equivalent

What is your current work status?

- Full time worker
- Unemployed/Housewife
- Retired
- Student
- Part-time worker/Intern

What is your current household income?

- Less than €1500 per month
- €1500 to €4999 per month

- ○ €5000 to €7999 per month
- ○ €8000 to €11999 per month
- ○ €12000+ per month

Powered by Qualtrics